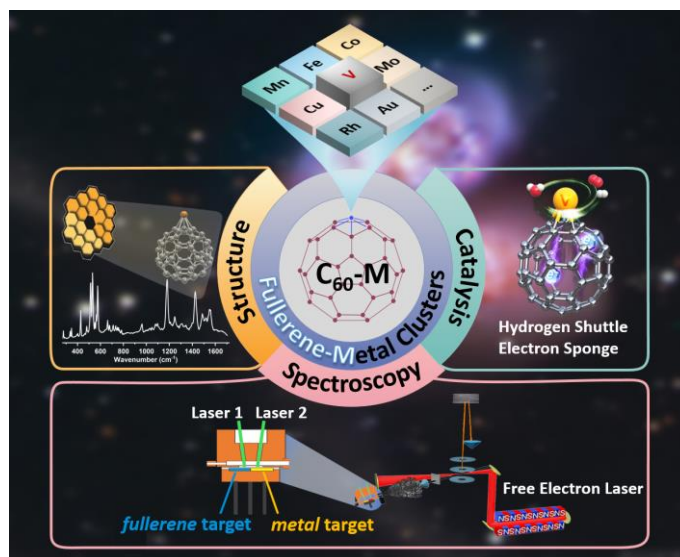


# Multifacets of Fullerene-metal Clusters: From Fundamental to Application

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## Conspectus

Buckminsterfullerene,  $C_{60}$ , was discovered through a prominent mass peak containing 60 atoms produced from laser vaporization of graphite, driven by Kroto's interest in understanding the formation mechanisms of carbon-containing molecules in space. Inspired by the geodesic dome-shaped architecture designed by Richard Buckminster Fuller, after whom the particle was named,  $C_{60}$  was found to have a football-shaped structure comprising of 20 hexagons and 12 pentagons. It sparked worldwide interest in understanding this new carbon allotrope, resulting in the awarding of the Noble Prize in Chemistry to Smalley, Kroto, and Curl in 1996.

Intrinsically,  $C_{60}$  is an exceptional species because of its high stability and electron-accepting ability, and its structural tunability by decorating or substituting either on its exterior surface or interior hollow cavity. For example, metal-decorated fullerene complexes have found important applications ranging from superconductivity, nanoscale electronic devices, organic photovoltaic cells to catalysis and biomedicine. Compared to the large body of studies on atoms and molecules encapsulated by  $C_{60}$ , studies on the exteriorly modified fullerenes, i.e., exohedral fullerenes, are scarcer. Surprisingly, to date uncertainty exists about a fundamental question: what is the preferable exterior binding site of different kinds of single atoms on the  $C_{60}$  surface?

In recent years, we have developed an experimental protocol to synthesize the desired fullerene-metal clusters and to measure their infrared spectra via messenger-tagged infrared multiple photon dissociation spectroscopy. With complementary quantum chemical calculations and molecular dynamics simulations,

we determined the most probable binding site of a metal, specifically a vanadium cation, on  $C_{60}$  is above a pentagonal center in a  $\eta^5$ -fashion. We explored the bonding nature between  $C_{60}$  and  $V^+$  and revealed that the high thermal stability of this cluster originates from large orbital and electrostatic interactions. Through comparing the measured infrared spectra of  $[C_{60}\text{-Metal}]^+$  with the observational Spitzer data of several fullerene-rich planetary nebulae, we proposed that the complexes formed by fullerene and cosmically abundant metals, for example iron, are promising carriers of astronomical unidentified spectroscopic features. This opens the door for a real consideration of Kroto's 30 year-old hypothesis that complexes involving cosmically abundant elements and  $C_{60}$  exhibit strong charge-transfer bands, similar to those of certain unidentified astrophysical spectroscopic features. We compiled a VibFullerene database and extracted a set of vibrational frequencies and intensities for fullerene derivatives to facilitate their potential detection by the James Webb Space Telescope. In addition, we showed that, upon infrared irradiation,  $C_{60}V^+$  can efficiently catalyze water splitting to generate  $H_2$ . This finding is attributed to the novel geometric-electronic effects of  $C_{60}$ , acting as "hydrogen shuttle" and "electron sponge", which illustrates the important role of carbon-based supports in single atom catalysts. Our work not only unveils the basic structures and bonding nature of fullerene-metal clusters, but also elucidates their potential importance in astrophysics, astrochemistry, and catalysis, showing the multifaceted character of this class of clusters. More exciting and interesting aspects of the fullerene-metal clusters, such as ultrafast charge transfer dynamics between fullerene and metal, and their relevance to designing hybrid fullerene-metal junctions for electronic devices, are awaiting exploration.

## KEY REFERENCES

- Xu, J.; Bakker, J. M.; Lushchikova, O. V.; Lievens, P.; Janssens, E.; Hou, G.-L. Pentagon, Hexagon, or Bridge? Identifying the Location of a Single Vanadium Cation on Buckminsterfullerene Surface. *J. Am. Chem. Soc.* **2023**, *145*, 22243.<sup>1</sup> This work reported high-quality infrared spectrum of  $C_{60}V^+$  via photofragmentation of Ar-tagged  $C_{60}V^+$ , and identified the most preferable binding site of  $V^+$  to be on top of a pentagon in a  $\eta^5$ -fashion with the aid of theoretical calculations and simulations.
- Hou, G.-L.; Lushchikova, O. V.; Bakker, J. M.; Lievens, P.; Decin, L. Buckyball-metal complexes as potential carriers of astronomical unidentified infrared emission bands. *Astrophys. J.* **2023**, *952*, 13.<sup>2</sup> This work reported laboratory infrared spectra of gas-phase fullerene-metal clusters,  $[C_{60}\text{-Metal}]^+$  (Metal = Fe and V), and comparison with Spitzer spectra from several fullerene-rich planetary nebulae suggests that fullerene-metal complexes could potentially contribute to the unidentified infrared spectroscopic features.
- Hou, G.-L.; Yang, T.; Li, M.; Vanbuel, J.; Lushchikova, O. V.; Ferrari, P.; Bakker, J. M.; Janssens, E. Water Splitting by  $C_{60}$ -Supported Vanadium Single Atoms. *Angew. Chem. Int. Ed.* **2021**, *60*, 27095-27101.<sup>3</sup> This work utilized  $C_{60}V^+$  as a single-atom-catalyst (SAC) model of carbon-based SACs for water splitting and found that  $C_{60}$  acts as a “hydrogen shuttle” to temporally store the transferred H atom and reduces the barrier to facilitate the  $H_2$  formation.
- Li, M.; Yang, T.; Bakker, J. M.; Janssens, E.; Hou, G.-L. Unveiling the role of  $C_{60}$ -supported vanadium single atoms for catalytic overall water splitting. *Cell Rep. Phys. Sci.* **2021**, *3*, 100910.<sup>4</sup> This work studied the overall water splitting mediated by  $C_{60}V^+$ , revealing that the vanadium center undergoes several formal oxidation state changes due to the unique electron donating and accepting ability of  $C_{60}$  and electronically acts as an “electron sponge”.

## 1. INTRODUCTION

In 1985, Smalley, Kroto, Curl, and co-workers published seminal work on the discovery of an all-carbon molecule that consists of sixty carbon atoms, i.e.,  $C_{60}$ .<sup>5</sup> This molecule was dominant in the mass spectra recorded for species formed from laser vaporized graphite, and was found to be highly stable, resistive to oxygen and chlorine gas etching as well as to 193 nm ultraviolet laser irradiation, suggesting that all carbon atoms in the molecule are saturated in chemical bonding.<sup>6</sup> Drawing inspiration from Richard Buckminster Fuller’s geodesic dome-shaped architecture and in consulting with mathematician William A. Veech, they proposed for  $C_{60}$  a football-shaped structure composing of 20 hexagons and 12 pentagons, now known as buckminsterfullerene.<sup>5,7</sup> The structure was not broadly accepted by the scientific community until the early 1990s when Kratschmer, Huffman, and co-workers succeeded in synthesizing  $C_{60}$  in large quantity and infrared spectroscopy and X-ray diffraction studies became feasible.<sup>8</sup>

Discovery of  $C_{60}$  marked a paradigm shift in carbon allotropes, permitting a molecular species of profound stability, high electron-accepting ability, and structural tunability.<sup>7</sup> Interest in encapsulating atoms, molecules, or clusters inside the hollow fullerene cavity, and chemical modification through decoration, substitution, or addition on the exterior surface to form exohedral fullerenes started almost together with the  $C_{60}$  discovery.<sup>9,10</sup> These novel materials, in particular metal-modified fullerenes, show interesting electronic, magnetic, and optical properties.<sup>11,12</sup> For instance, alkali and alkaline earth metal-doped fullerenes present high hydrogen storage capability,<sup>13,14</sup> alkali metal-doped fullerene films show superconducting behavior,<sup>15</sup> transition metal-doped fullerenes have interesting applications in organic

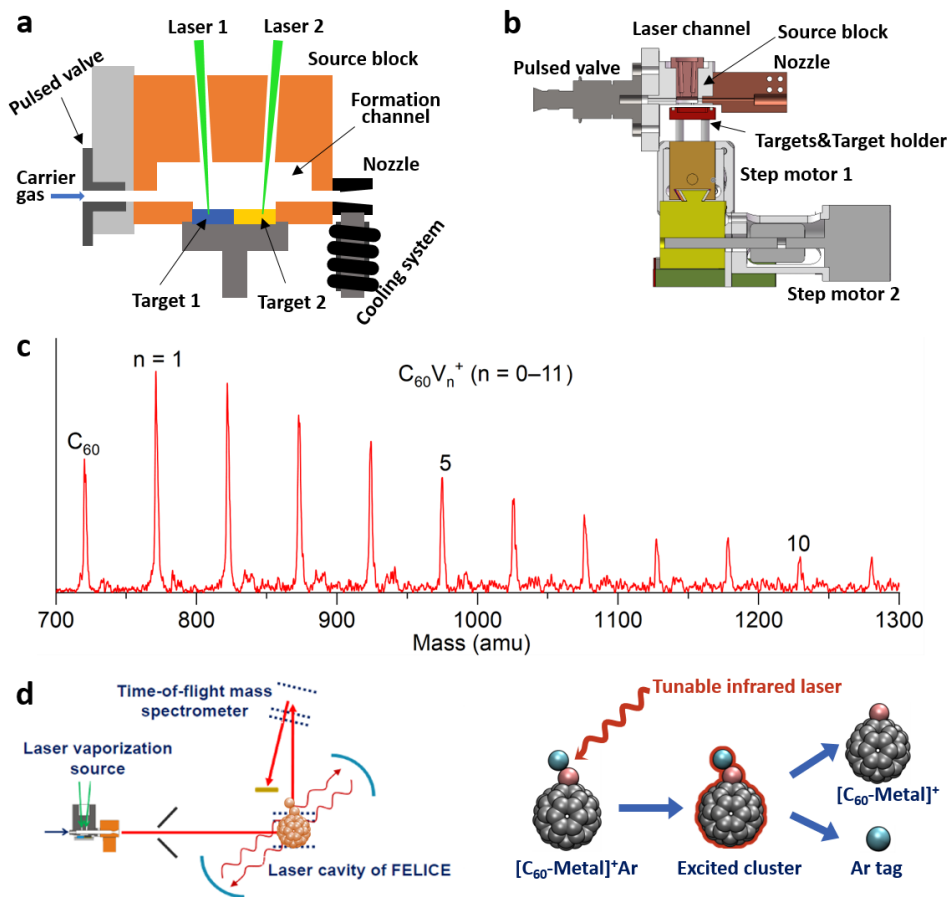
photovoltaic cells,<sup>12,16</sup> and the heterojunction structures of fullerene with coinage metals have the potential to be used in nanoscale electronic devices.<sup>17</sup> All these phenomena depend on the chemical bonding and interaction nature between fullerene and metal, emphasizing the importance to understand fullerene-metal hybrid structures.<sup>12,18-20</sup>

Reminiscing the original motivation of Kroto, Smalley, Curl, and co-workers that led to the discovery of C<sub>60</sub>, they already wrote in the 1985 Nature paper: “Even more speculatively, C<sub>60</sub> or a derivative might be the carrier of the diffuse interstellar lines.”<sup>5</sup> This seemingly bold statement at that time, together with Kroto’s further speculation in 1992,<sup>21</sup> motivated us for further research in this direction. In fact, C<sub>60</sub> has now been identified unequivocally in a variety of astrophysical environments, contributing in neutral form to observed infrared vibrational features at 7.0, 8.5, 17.4, and 18.9  $\mu\text{m}$ ,<sup>22,23</sup> and in cationic form to the near infrared diffuse interstellar bands (DIBs) at 9348, 9365, 9428, 9577, and 9632 Å.<sup>24,25</sup> Still, there remain many unidentified observational features in both infrared and optical domains.<sup>26,27</sup> Then an interesting question arises: Could fullerene derivatives, like fullerene-metal complexes, exist in space and contribute to unidentified astrophysical spectroscopic features?

A prerequisite to this question is to obtain their high-quality laboratory spectra as to compare with the astronomical data. In this Account, we review the current fundamental knowledge in respect to the above questions, with focus on our recent efforts on investigating the spectroscopy, geometries, and bonding properties of fullerene-metal clusters, as well as their astrophysical relevance. We also describe novel geometric and electronic effects of fullerenes in catalysis that we found when employing fullerene-supported metal atoms and clusters as models of single atom catalysts (SACs) and/or single cluster catalysts (SCCs), with C<sub>60</sub> acting as a carbon-based support. Finally, we present our perspective on these fascinating clusters that are not only interesting objects of fundamental research, but also useful models for problems associated with application fields, such as catalysis, ultrafast charge transfer dynamics, and novel materials based on fullerene-metal junctions.

## 2. SYNTHESIS, MASS SPECTROMETRY, AND SPECTROSCOPY OF FULLERENE-METAL CLUSTERS

Synthesis of fullerene-metal clusters with controllable composition and size is a crucial first step to study their spectroscopic features and structures. In our work, the desired fullerene-metal clusters are produced using a dual-laser ablation source, whose design is based on the one that was used in the discovery of C<sub>60</sub>.<sup>1-5</sup> Figures 1a and 1b show a scheme of the cluster source utilized in Leuven, and of a similar source employed in Xi'an. Two vacuum stepper motors drive a holder with two rectangular targets, each having dimensions  $\sim 7\text{ mm} \times 25\text{ mm}$ , in a rectangular loop. A spring presses the Teflon-sealed target holder against the source block. The cluster forms in a cylindrical channel (17 mm long, 3 mm diameter) which is directly connected with a 10 Hz pulsed supersonic gas valve that introduces high-purity inert carrier gas. The ablation lasers hit the targets through two small channels that are nearly orthogonal to the formation channel, as indicated by arrowed green bars in Fig. 1a. The source block is thermally isolated from the pulsed valve by a 15 mm thick Teflon plate.



**Figure 1.** Scheme of the dual-target dual-laser ablation sources in (a) Leuven and (b) Xi'an. (c) Mass spectrum of  $C_{60}V_n^+$  ( $n = 0-11$ ) clusters. (d) Overview of the IRMPD technique. Part of the figure modified from ref. 28. Copyright 2023 Elsevier.

Two Nd:YAG laser beams (532 nm, 10 Hz), whose energy densities are controlled by focal lenses and the time delays between flash lamp and Q-switch, are focused on the targets. For the fullerene target, one has to finely adjust the focal position and laser energy density to avoid the fragmentation of fullerene. The laser ablated metal plasma containing metal cations and the vaporized neutral  $C_{60}$  molecules collide with each other in the presence of He gas that initiates the formation and cooling of the formed clusters. In the extension tube that follows the cluster formation channel, nozzles with different geometries can be mounted for flexible control of the expansion conditions and thereby of the cluster size distribution. Furthermore, through resistive heating and liquid nitrogen cooling of the cluster source, the temperature of clusters can be controlled from room temperature down to 100 K, facilitating cluster temperature-dependent reactivity studies.<sup>29</sup> Then, a skimmer and a slit aperture both at 2 mm are used to select the central portion of the generated cluster beam, before it enters the extraction field of a reflectron time-of-flight (TOF) mass spectrometer, where the clusters are separated based on mass-to-charge ratio and detected.

In the gas phase, the number densities of clusters typically are in the range from a few hundreds to a few ten thousands particles per cubic centimeter, which are too low for conventional infrared absorption spectroscopy that measures the attenuation of light as a function of light frequency.<sup>30</sup>

Therefore, infrared action spectroscopy, more specifically infrared multiple photon dissociation (IRMPD) spectroscopy is employed, which probes mass spectrometrically the absorption of light by monitoring the changes of the cluster signals upon exposure to the infrared light.

The action spectroscopy described in this Account, illustrated in Figure 1d, is conducted with the intense far- to mid-infrared light from the Free Electron Laser for Intra-Cavity Experiments (FELICE) at the Free Electron Lasers for Infrared eXperiments (FELIX) facility of the Radboud University (Nijmegen, the Netherlands). Infrared light from FELICE is overlapped with the cluster beam to perform IRMPD experiments.<sup>31</sup> Absorption of multiple infrared photons, in resonance with a specific vibrational mode, and intra-molecular vibrational redistribution heats up the clusters. When the internal energy of the messenger-tagged fullerene-metal cluster is high enough, desorption of messenger from the cluster occurs on the timescale of the experiment (Figure 1d), which can be probed sensitively by time-of-flight mass spectrometry. For more details about IRMPD spectroscopy, we refer to a comprehensive recent review by Fielicke.<sup>30</sup>

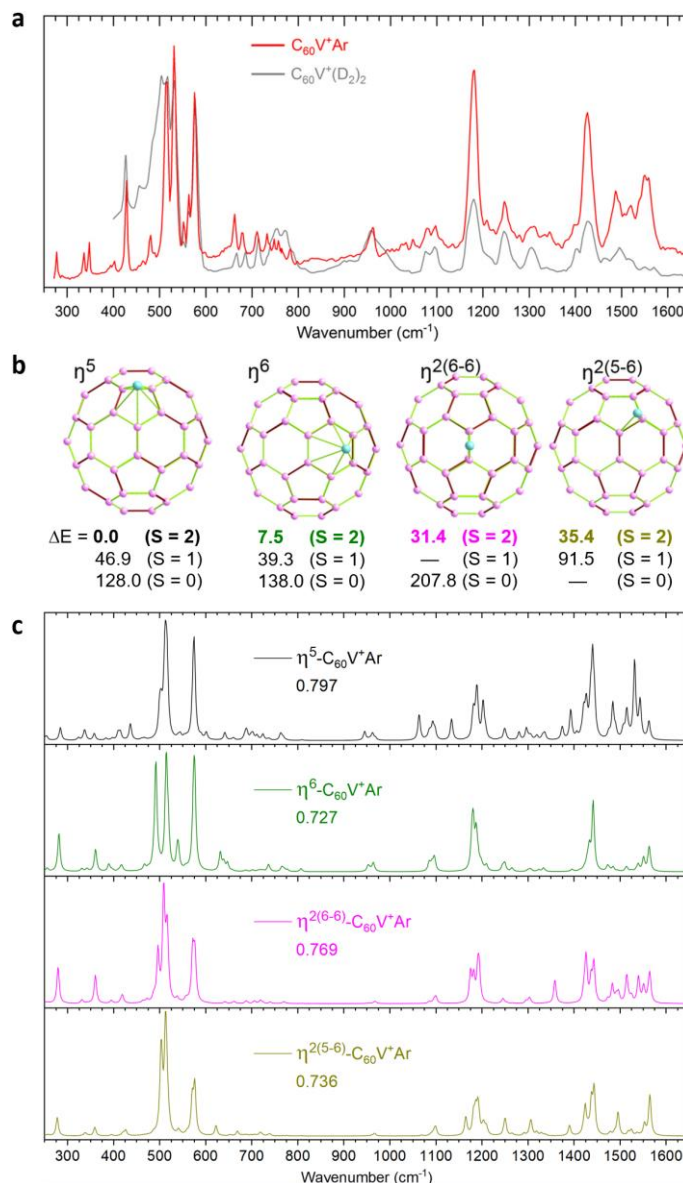
### 3. STRUCTURES AND CHEMICAL BONDING OF FULLERENE-METAL CLUSTERS

$C_{60}$  has an  $I_h$  symmetry, implying there are five possible exohedral binding sites, i.e., atop of a carbon atom  $\eta^1$ , a pentagon center  $\eta^5$ , a hexagon center  $\eta^6$ , a bridge between a pentagon and a hexagon  $\eta^{2(5-6)}$ , and a bridge between two hexagons  $\eta^{2(6-6)}$ . Knowledge about the preferred binding site is crucial to unravel the nature of the metal-fullerene bonds, and such knowledge will in turn allow to deeper understand and/or improve the design of fullerene-based functional materials. To date, there is little experimental structural benchmarking data. In this regard, mostly quantum chemical calculations are used, but they do not allow for conclusive results, because the energetic ordering of the isomers may depend on the theoretical methods, and the energy differences between different isomers may fall within their typical accuracy.

#### 3.1. Single Vanadium Cation on $C_{60}$ surface

Infrared spectra of  $C_{60}V^+$  were measured via photodissociation of Ar- and  $D_2$ -tagged  $C_{60}V^+$  complexes using the messenger-tagging IRMPD technique (Figure 2a). It can be seen that if one disregards the peak intensities and band features corresponding to motions of  $D_2$ -tag in the 400–500  $cm^{-1}$  range and around 750  $cm^{-1}$ , the two spectra are almost identical. However, the spectrum using an Ar-tag shows a better resolution with sharper features than the  $D_2$ -tagged one. Because the binding strength of Ar with  $C_{60}V^+$  is much smaller than that of  $D_2$ , less photons have to be absorbed to induce Ar desorption, thus less heating up of the complex.

Figure 2b presents the four optimized isomers of  $C_{60}V^+$ , including  $\eta^5$ ,  $\eta^6$ ,  $\eta^{2(6-6)}$ , and  $\eta^{2(5-6)}$  together with their relative stabilities. The  $\eta^1$  structure was not stable and converged to either  $\eta^5$  or  $\eta^6$ ; the theoretical calculations show that high spin states are at least 30 kJ/mol more stable than low spin states. From energetics, the  $\eta^{2(6-6)}$  and  $\eta^{2(6-5)}$  bridge structures are ruled out as they are >30 kJ/mol less stable than the  $\eta^5$  structure.



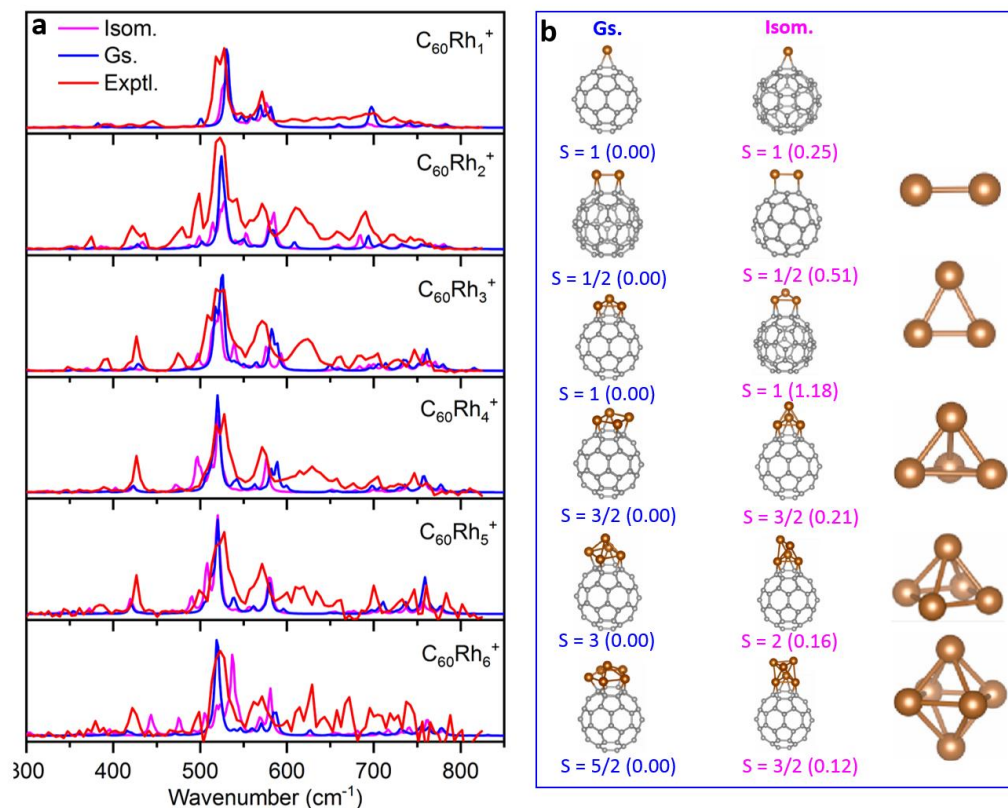
**Figure 2.** (a) IRMPD spectra of  $D_2$ - and Ar-tagged  $C_{60}V^+$ . (b) Four optimized structures and their relative stabilities ( $\Delta E$ , in kJ/mol) at BPW91/6-31G(d) level. (c) Simulated spectra of those isomers with  $S=2$  spin states. Cosine similarity scores with the Ar-tagged experimental spectrum are indicated for each simulated one. The simulated spectra are convolved using Lorentzian line shapes of 6  $cm^{-1}$  fwhm. Figure modified from ref. 1. Copyright 2023 American Chemical Society.

Given that the typical accuracy of density functional theory (DFT) calculations is about 10 kJ/mol, we cannot unambiguously assign either  $\eta^5$  or  $\eta^6$  structure based on the calculated energy difference of 7.5 kJ/mol. It seems challenging to draw a conclusive assignment of the Ar-tagged experimental spectrum in Figure 2a to any of the isomers solely based on visualization. For instance, the  $\eta^6$  simulated spectrum matches quite good in the 900–1600  $cm^{-1}$  range except for a missing band just below 1500  $cm^{-1}$ , but more discrepancies show up below 800  $cm^{-1}$  and particularly below 450  $cm^{-1}$ . The  $\eta^5$  structure overall matches the experiment, but it fails to accurately reproduce the relatively large splitting of the experimental feature around 510–530  $cm^{-1}$ . Hence, the cosine similarity score, an objective measure to reflect the

similarity between two curves,<sup>32</sup> are employed to quantify the agreement between experimental and calculated infrared spectra. It is found that the cosine similarity score is highest (0.797) for the  $\eta^5$  structure, allowing to assign  $\eta^5$   $C_{60}V^+$  the most probable structure in the experiment. This deviates from previous theoretical predictions,<sup>20,33</sup> and the empirical interpretation employing the 18-electron counting rule that favored the  $\eta^6$  structure.<sup>34</sup>

We also calculated the isomerization pathways of four  $C_{60}V^+$  isomers and conducted Born-Oppenheimer molecular dynamics (BOMD) simulations of  $\eta^5$  and  $\eta^6$   $C_{60}V^+$ . We found no interconversion between the different isomers of  $C_{60}V^+$  below 800 K, while at 1000 K the interconversion starting from  $\eta^6$   $C_{60}V^+$  leads to  $\eta^5$ , consistent with the slightly higher stability of  $\eta^5$   $C_{60}V^+$ . Energy decomposition analysis combined with natural orbitals for chemical valence revealed that electrostatic ( $\Delta E_{\text{elstat}}$ ) and orbital ( $\Delta E_{\text{orb}}$ ) interactions are the main contributors to the stability of  $\eta^5$   $C_{60}V^+$ . Orbital interaction mainly involves  $C_{60}$  and the five d orbitals of  $V^+$  with clear charge transfer between  $V^+$  and  $C_{60}$ , but the exact contribution of each d orbital depends on the specific isomer.

### 3.2. Rhodium cluster cations on $C_{60}$ surface



**Figure 3.** (a) Comparison between calculated  $C_{60}Rh_n^+$  and measured  $D_2$ -tagged  $C_{60}Rh_n^+$  ( $n = 1-6$ ) spectra. (b) Optimized  $C_{60}Rh_n^+$  and  $Rh_n^+$  structures with PBE functional in Quantum-ESPRESSO software. The projected augmented wave (PAW) method was applied to model the electron-ion core interactions. The number of explicitly treated electrons is 17 for Rh (4s24p64d85s1), four for C (2s22p2), and one for H. Spin states and relative energies (in eV) are given. Figure modified from ref. 35. Copyright 2022 Elsevier.

Figure 3 shows the measured IRMPD spectra of  $C_{60}Rh_n^+$  ( $n = 1-6$ ) clusters via  $D_2$ -tagged fullerene-rhodium complexes, and their comparison with the simulated infrared spectra of the ground state (Gs.)



and first low-lying isomeric (Isom.) structures. It can be seen that in general all spectra show two intense bands around 520 and 570  $\text{cm}^{-1}$ , which are vibrational signatures of neutral  $\text{C}_{60}$ , as perturbed by the adsorbed rhodium and hydrogen. Broader bands are observed in the 600–750  $\text{cm}^{-1}$  range, which turn out to be features mainly arising from the presence of  $\text{D}_2$ -messenger molecules. Since Rh–Rh vibrational modes have much lower frequencies, their signatures are not observed in the 300–850  $\text{cm}^{-1}$  range.<sup>36</sup> All the other features are mainly associated with modes of the adsorbed metal and hydrogen perturbed  $\text{C}_{60}$ -cage modes, which were infrared inactive in bare  $\text{C}_{60}$  due to the  $I_h$  symmetry. Overall, the main features in the IRMPD spectra are reasonably reproduced by the calculated spectra for the Gs. structures, except in the 600–750  $\text{cm}^{-1}$  region. Comparison of calculated spectra of untagged and deuterated  $\text{C}_{60}\text{Rh}_n^+$  clusters reveals that deuteration leads to the extra bands in this region.

In the case of a single adsorbed  $\text{Rh}^+$  atom, it is found that  $\eta^{2(6-6)}$  site is energetically most favorable. For  $n = 2-6$ , the  $\text{Rh}_n^+$  clusters preferentially bind to a  $\text{C}_{60}$  hexagon, likely due to the smaller area of a pentagon compared to a hexagon. The rhodium clusters on  $\text{C}_{60}$  are three-dimensional for  $n \geq 5$ , with  $\text{Rh}_5^+$  and  $\text{Rh}_6^+$  being pyramids with square and pentagonal bases; while free rhodium clusters become three-dimensional at  $n = 4$  with a tetrahedral shape. That means the  $\text{C}_{60}$  support changes the geometry of  $\text{Rh}_4^+$ , which is also found for  $\text{Rh}_6^+$ . It seems that fullerene supported  $\text{Rh}_n^+$  clusters tend to maximize the number of Rh atoms that are in direct contact with the fullerene, in contrast to the case of cobalt clusters on  $\text{C}_{60}$ ,<sup>37</sup> where maximally three Co atoms are in direct contact with the fullerene. That difference may arise from a delicate balance between the strength of metal-fullerene or metal-carbon and metal-metal interactions, emphasizing that care must be taken when investigating the structural evolution of different metals, and thus their properties and applications. Besides, we also studied the binding locations of one or two boron atoms inside or outside the  $\text{C}_{60}$  cage using DFT calculations, and refer interested readers to ref. 38.

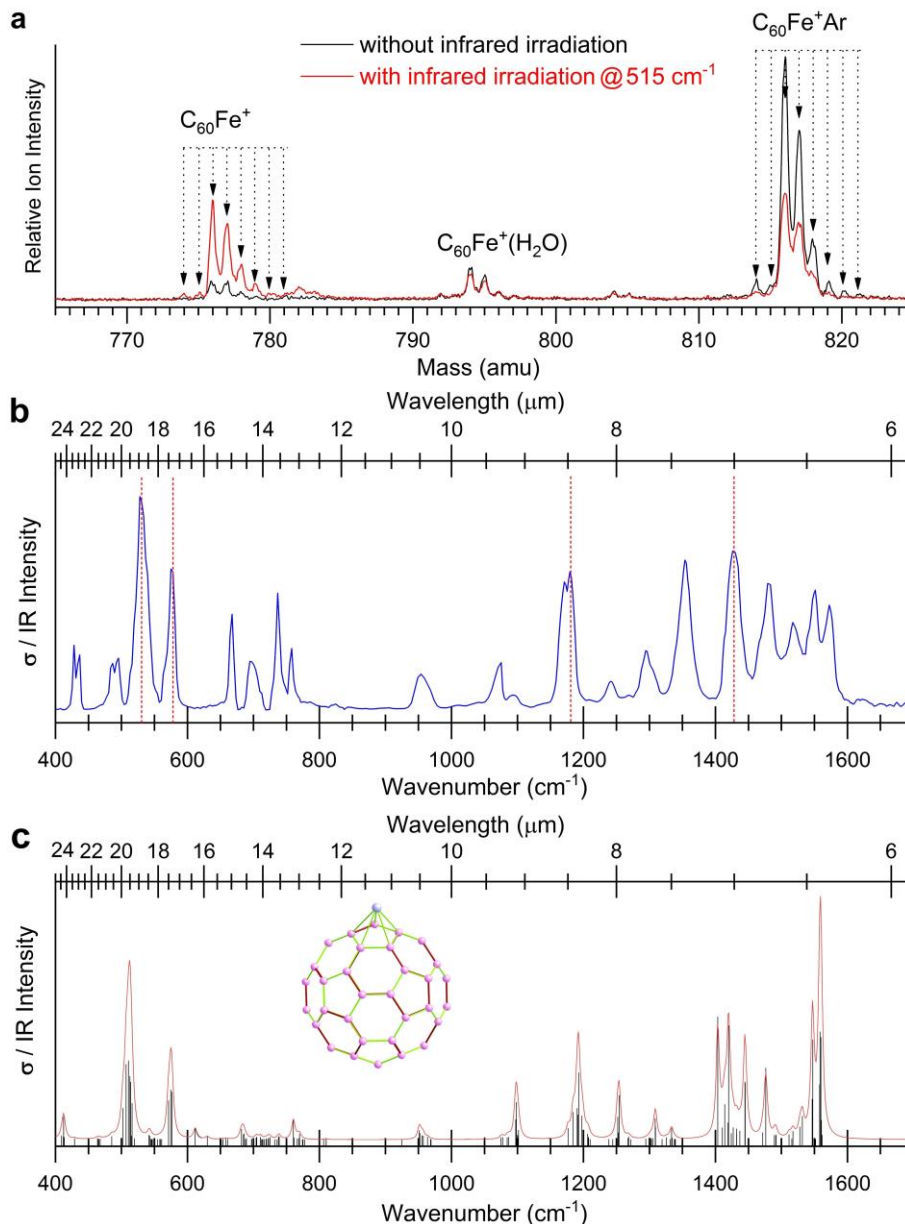
#### 4. ASTROPHYSICAL RELEVANCE AND IMPLICATIONS OF FULLERENE-METAL COMPLEXES

Detecting complex organic molecules in space has been one big challenge for astrophysics.<sup>39</sup> To date, over 300 species, composed of different chemical elements, have been identified in a variety of astrophysical environments, including protoplanetary disks, interstellar and circumstellar media, and exoplanets, with the  $\text{C}_{60}$ ,  $\text{C}_{60}^+$ , and  $\text{C}_{70}$  the largest molecules so far detected.<sup>22-24,39,40</sup> The presence in space of small metal-containing organic molecules, such as  $\text{MgNC}$  or  $\text{CaCN}$ ,<sup>40,41</sup> motivated the idea that the interaction of cosmically abundant metals with  $\text{C}_{60}$  may form fullerene-metal complexes. In certain astrophysical environments both ingredients for fullerene-metal complex formation, i.e., metals and fullerenes, are present. In fact, already in their 1985 seminal paper, Kroto et al. speculated that  $\text{C}_{60}$  or its derivative might be the carrier of diffuse interstellar lines.<sup>5</sup> Fullerene-derived molecules recently have been proposed as potential candidates to explain unidentified infrared emission (UIE) bands, the UV-bump and DIBs.<sup>22,24,39,42</sup>

##### 4.1. Charge-transfer fullerene-metal complexes as promising carriers of UIE bands

Figure 4 presents a mass distribution of synthesized  $\text{C}_{60}\text{Fe}^+$  complexes and the IRMPD spectrum of Ar-tagged  $\text{C}_{60}\text{Fe}^+$ , as well as the DFT simulated infrared spectrum of  $\text{C}_{60}\text{Fe}^+$ . Except for the four main band features coinciding with the neutral  $\text{C}_{60}$  bands (vertical red lines in Figure 4b), many more vibrational bands appear. Theoretical calculations in Figure 4c essentially reproduce all the observed features in Figure 4b, except that the intensities of a few bands are slightly off. Such intensity mismatch is common in the comparison of quantum chemical calculations and IRMPD experiments because of the multiple-photon nature of the experiment.<sup>43</sup> Those findings lend confidence for the reliable prediction of the infrared spectra of  $\text{C}_{60}$ -metal complexes with metals being Li, Na, K, Mg, Ca, and Al,<sup>42</sup> which are cosmically

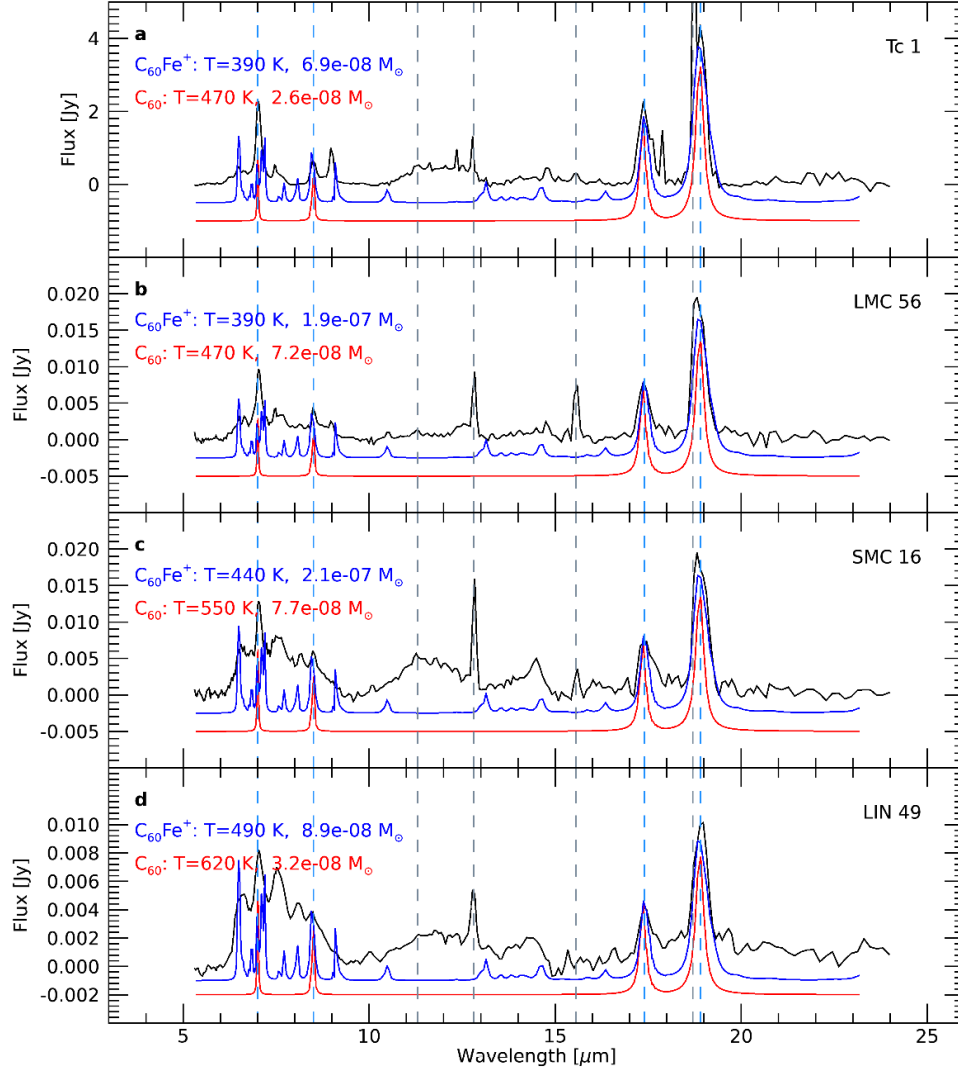
abundant (Figure A3 in ref. 2). Quantum chemical calculations show that all these fullerene-metal complexes possess similar infrared spectra as  $C_{60}Fe^+$ . Detailed analysis of the vibrational modes indicate that most of the infrared features involve the  $C_{60}$  cage motions, perturbed by the attached atom. Hence, we are convinced that the laboratory infrared spectrum of  $C_{60}Fe^+$  is prototypical for various  $[C_{60}\text{-Metal}]^+$  complexes.



**Figure 4.** Laboratory synthesis and infrared spectroscopy of  $C_{60}Fe^+$ . (a) Mass distributions of  $C_{60}Fe^+$  and its Ar-tagged complexes. (b) IRMPD spectrum of Ar-tagged  $C_{60}Fe^+$ . (c) Simulated spectrum of  $C_{60}Fe^+$  at the BPW91/6-31G(d) level of theory with its calculated structure. The theoretical intensities are plotted with sticks and convolved using Lorentzian line shapes of 6 cm<sup>-1</sup> fwhm. Figure modified from ref. 2. Copyright 2023 American Astronomical Society.

We compared the laboratory infrared spectrum of  $C_{60}Fe^+$  with the Spitzer spectra of several fullerene-rich planetary nebulae (PNe) and found a strong positive linear cross-correlation. The co-existence of different  $[C_{60}\text{-metal}]^+$  complexes in the same astronomical object can slightly affect the band positions

and band intensities. That may explain the observational shoulders and asymmetric band profiles observed in the 17–20  $\mu\text{m}$  region, and regulate the band ratios of the four neutral  $\text{C}_{60}$  bands by the varied spectral intensities for different metals (Figure 5). Such observation could potentially resolve the long-standing question about the wide-spread astronomical intensity ratios of the four neutral  $\text{C}_{60}$  bands.<sup>2</sup> Furthermore, we found that the infrared spectra of fullerene-metal complexes may also explain several yet unidentified or poorly explained features in the observational spectra.



**Figure 5.** Comparison between the observational Spitzer spectra of four fullerene-rich PNe (black) and the thermal emission model for  $\text{C}_{60}\text{Fe}^+$  (blue) and  $\text{C}_{60}$  (red). (a) Tc 1, (b) LMC 56, (c) SMC 16, (d) Lin49. The excitation temperatures and the required mass derived from the thermal emission model based on a microcanonical ensemble are provided in each panel. Blue and gray dashed lines indicate the four observational or calculated  $\text{C}_{60}$  bands and strong atomic emission lines, respectively. Figure modified from ref. 2. Copyright 2023 American Astronomical Society.

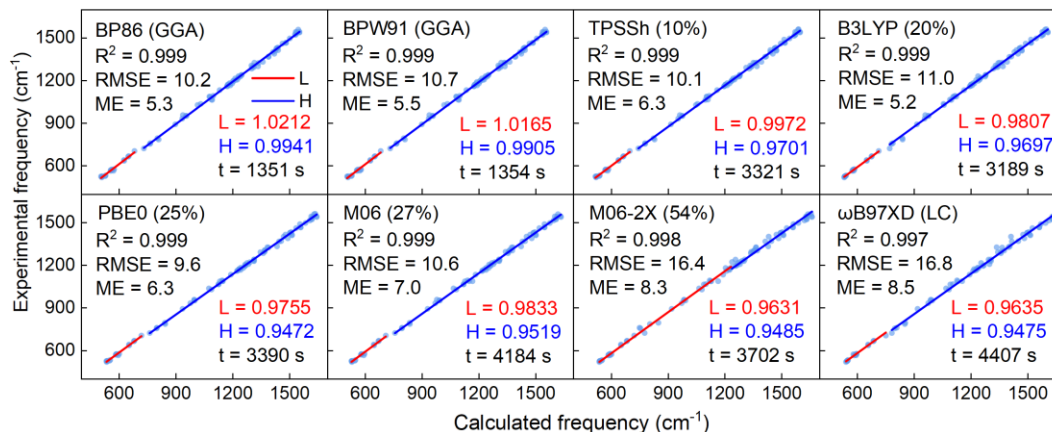
Figure 5 shows the temperature and the abundance required for each species to reproduce the observed infrared emission spectra of the four fullerene-rich PNe, based on a thermal emission model which treats the fullerene-metal complexes as a microcanonical ensemble. About  $2.6\text{--}7.7 \times 10^{-8}$  solar mass ( $M_{\odot}$ ) of pure  $\text{C}_{60}$  is required to reproduce the four major emission bands. For  $\text{C}_{60}\text{Fe}^+$ , about  $\sim 0.69\text{--}2.1 \times 10^{-7} M_{\odot}$  of  $\text{C}_{60}\text{Fe}^+$  or  $\sim 1.9\text{--}6.1\%$  of the available carbon with excitation temperatures above  $\sim 400$  K can give

an equally good fit, in addition to the fact that they could also account for several unexplained infrared emission features. Since the ionization potentials of both  $C_{60}$  and  $C_{60}Fe$  are low, their ionized form could be favored. However, due to the complicated charge balance, i.e., competition among ionization, electron attachment, and recombination events in combination with a lack of parameters associated with those processes, it is difficult if not impossible to accurately estimate the fractions.

Based on the good fits of the thermal emission model and high calculated cross-correlation coefficients, we proposed that gaseous ionized fullerene-metal complexes, in addition to neutral  $C_{60}$ , are promising candidates for explaining the infrared emission spectra of the fullerene-rich PNe. Such results open the door for a real consideration of Kroto's over 30-year old hypothesis.<sup>21</sup> Moreover, the complexation of metals with fullerene could result in a much richer spectrum in the visible to near-infrared spectral range than bare fullerene exhibits (see Figure A13 in ref. 2), due to the charge transfer from metal to carbon cage. That makes  $[C_{60}\text{-metal}]$  complexes also potential candidates for DIBs. It is known that metals are active catalysts for the synthetic production of various carbon nanostructures, including carbon cages even under oxygen- and hydrogen-rich conditions,<sup>44,45</sup> and might play a catalytic role in fullerene formation in space. Consideration of their role in cosmic chemical transformations might lead to a breakthrough in our understanding of cosmic carbon chemistry.

## 4.2. Construction of the VibFullerene database

Motivated by the conjecture that fullerenes can undergo complicated chemical processes during stellar evolution, forming fullerene derivatives like hydrogenated fullerenes, fullerene-metal, and fullerene-hydrocarbon complexes, and given the difficulty to measure infrared spectra of these large species, a VibFullerene database was compiled that includes all so far measured vibrational frequencies of fullerene-related species, including  $C_{60}$ ,  $C_{70}$ ,  $C_{60}^+$ ,  $C_{60}O^+$ ,  $C_{60}OH^+$ ,  $C_{60}H^+$ , and  $C_{70}H^+$ , totally 92 frequencies.<sup>46</sup> A systematic theoretical study on these species was conducted using several DFT methods, deriving a set of range-specific scaling factors for vibrational frequencies following a least-square fitting procedure (Figure 6).



**Figure 6.** Scaling factors for the low- and high-frequency (L and H) ranges of different functionals with 6-31G\*\* basis set. The HF component in each functional is given in parenthesis. Root mean square error (RMSE) median error (ME) for the fittings, and time cost (t) for the frequency analysis of  $C_{60}$  on a 52-core SE24HIJTP workstation are provided. Modified from ref. 46. Copyright 2023 Royal Astronomical Society.

Figure 6 shows that by separating the infrared spectra into low- and high-frequency regions, satisfactory fittings with  $R^2$  values larger than 0.997 can be obtained for all the investigated functionals. It is found that the functionals with high Hartree-Fock (HF) components show tendency to overestimate the

calculated harmonic frequencies, necessitating small scaling factors. Overall, the functionals with low-HF components seem to exhibit smaller root mean square error (RMSE) and median error values, showing slightly better performance. As such, the generalized gradient approximations (GGA), i.e., BP86 and BPW91, present a good balance between the computational accuracy and efficiency, and are thus suitable choices for simulating the infrared spectra of fullerene-related species.

To facilitate potential future search, analysis, and interpretation of these fullerene-derived species in space by the James Webb Space Telescope (JWST), which has high-resolution and high-sensitivity in the 4.9–28.3  $\mu\text{m}$  spectral range, we listed all the scaled harmonic frequencies and corresponding transition intensities of these fullerene derivatives for their potential detection (see Table 4 in ref. 46). Definitive confirmation of a specific species in space via these frequencies, however, is challenging because of the potential coincidence of infrared features from different species in addition to the possibility that some band features are hidden within broad infrared emission plateaus. Nevertheless, the whole set of vibrational modes and their frequencies should be considered to understand its cosmic relevance, rather than focusing on a specific band feature.<sup>2</sup> The in-depth understanding on the distinctive infrared signatures of these carbon species is expected to contribute to our knowledge about the organic inventory and carbon evolution in the universe.

## 5. CATALYTIC IMPLICATIONS OF FULLERENE-METAL CLUSTERS

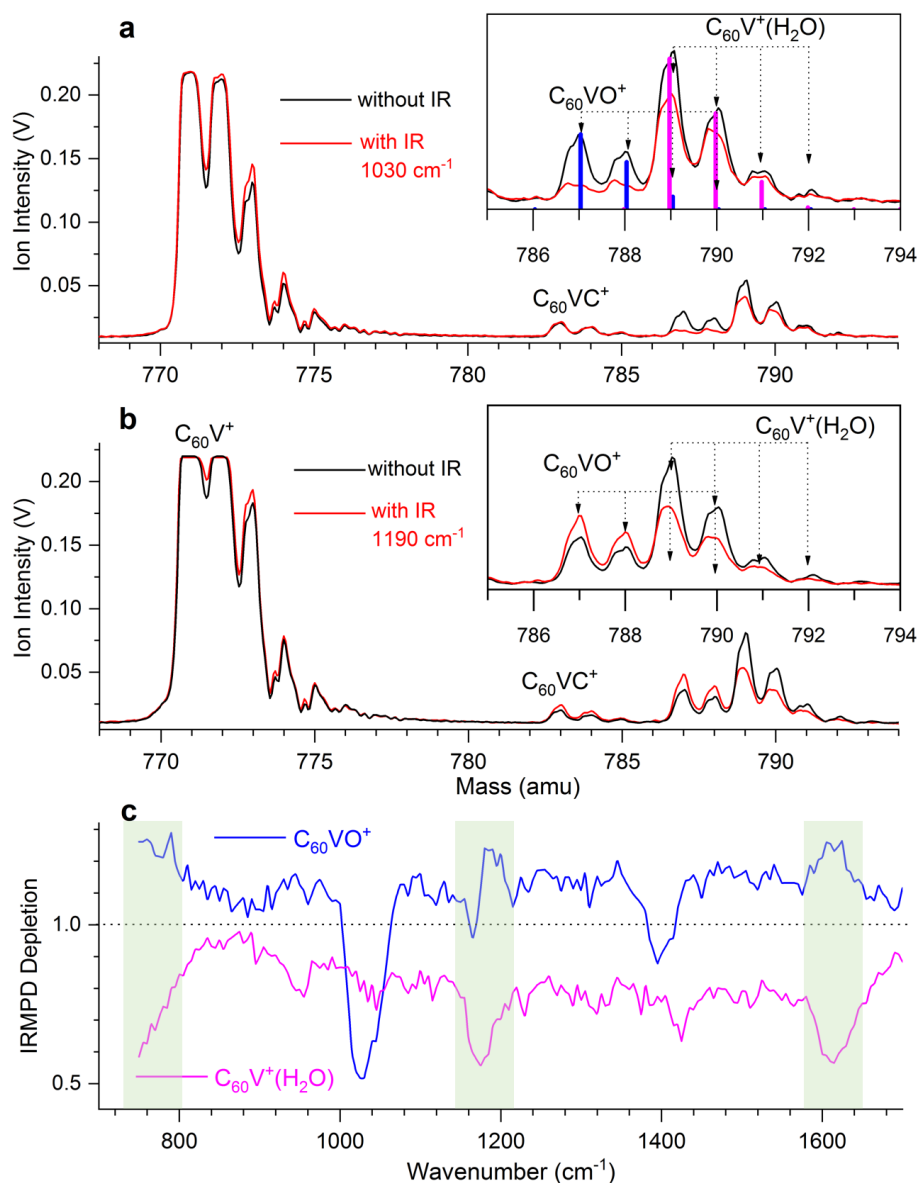
Fullerene-metal clusters are reminiscent of the fast developing SACs and SCCs, with isolated metal atoms or clusters dispersed on a support.<sup>47</sup> Since  $\text{C}_{60}$  has a high mechanical stability and a relatively large size of 0.71 nm, it can be regarded as a model for porous carbon materials, such as graphite, graphene, and carbon nanotubes. Fullerene-metal clusters can be regarded as simplified model of SACs and SCCs for investigating the synergy of metal-support and metal-metal interactions in heterogeneous catalysis. We focused on one of the most prototype catalytic reactions: water splitting to generate hydrogen, which is a promising approach to produce clean energy for combating global warming and energy crisis.

### 5.1 Fullerene as a “hydrogen shuttle” for water splitting: geometric effect

In an infrared spectroscopic study of fullerene-vanadium clusters, it was found that  $\text{C}_{60}\text{V}^+$  can catalyze water splitting under the infrared irradiation,  $\text{C}_{60}\text{V}^+ + \text{H}_2\text{O} \rightarrow \text{C}_{60}\text{V}^+(\text{H}_2\text{O}) \rightarrow \text{C}_{60}\text{VO}^+ + \text{H}_2$ ; a reaction that was not observed on isolated  $\text{V}^+$  without  $\text{C}_{60}$ .<sup>3,48</sup> The entrance channel adduct  $\text{C}_{60}\text{V}^+(\text{H}_2\text{O})$  and possible reaction product  $\text{C}_{60}\text{VO}^+$  were characterized with IRMPD spectroscopy.

Figure 7 shows that upon irradiation with 1030  $\text{cm}^{-1}$  infrared laser light, both  $\text{C}_{60}\text{V}^+(\text{H}_2\text{O})$  and  $\text{C}_{60}\text{VO}^+$  decrease in intensity; while at 1190  $\text{cm}^{-1}$ ,  $\text{C}_{60}\text{V}^+(\text{H}_2\text{O})$  decreases and  $\text{C}_{60}\text{VO}^+$  increases. That suggests infrared-induced water splitting of  $\text{C}_{60}\text{V}^+(\text{H}_2\text{O})$  via  $\text{H}_2$  loss at 1190  $\text{cm}^{-1}$ . The calculated dissociation energy (66 kJ/mol) is 50 kJ/mol lower than the energy required for the anticipated water loss channel.

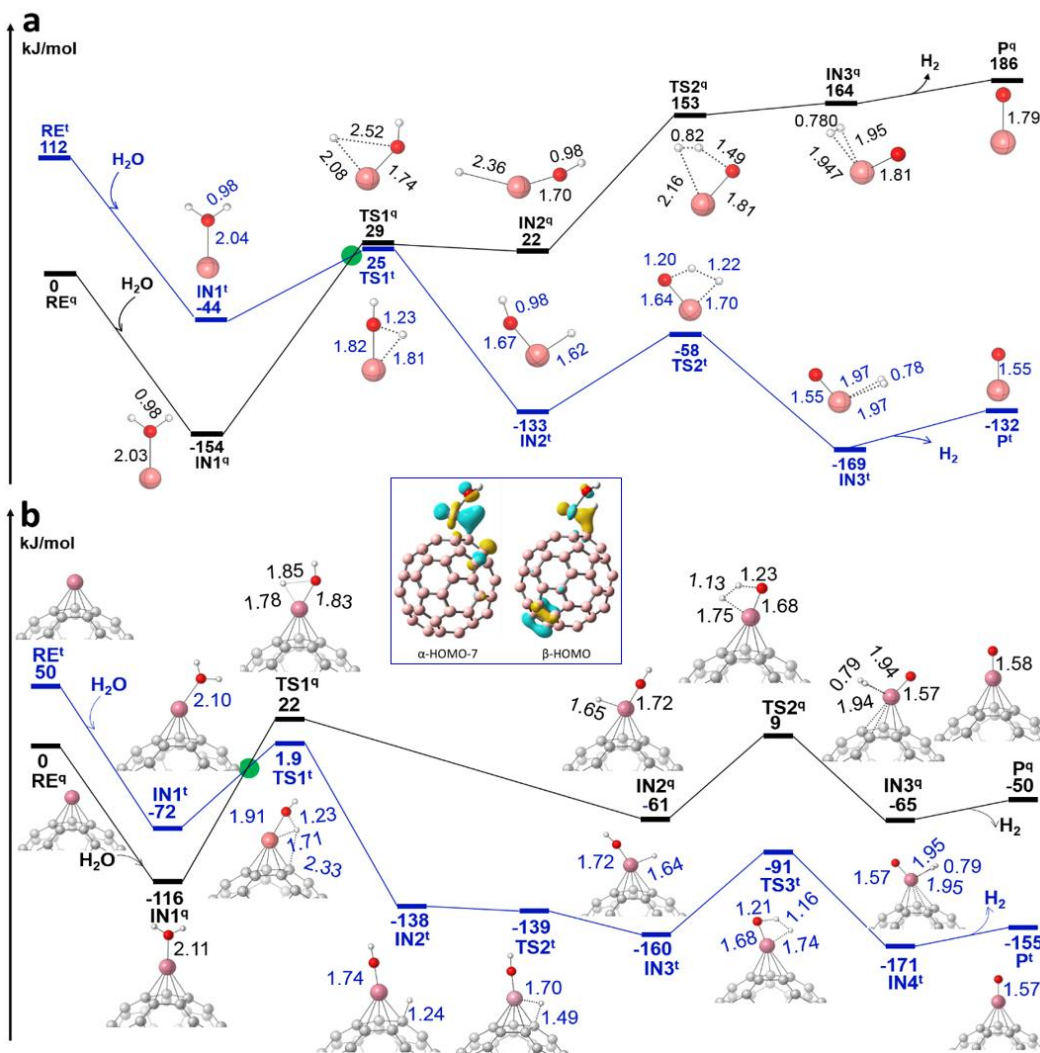
Figure 7c points out that  $\text{H}_2$ -loss channel also occurs at other vibrational frequencies, particularly around the typical water bending frequency of 1620  $\text{cm}^{-1}$ . Simultaneous signal loss of  $\text{C}_{60}\text{V}^+(\text{H}_2\text{O})$  and signal increase of  $\text{C}_{60}\text{VO}^+$  implies that the  $\text{H}_2$ -loss and  $\text{H}_2\text{O}$ -loss are competing. The calculated  $\text{H}_2$ -loss channel ratio with infrared irradiation at 1620  $\text{cm}^{-1}$  is 15%, 45% at 1190  $\text{cm}^{-1}$ , and 25% at 760  $\text{cm}^{-1}$ , suggesting mode-specific photocatalytic water splitting. Since the  $\text{H}_2$ -loss channel was not observed in previous infrared dissociation experiments on  $\text{V}^+(\text{H}_2\text{O})$  without  $\text{C}_{60}$ ,<sup>48</sup> it is intriguing to understand what is the role of  $\text{C}_{60}$ .



**Figure 7. (a–b)** Mass distributions of  $C_{60}V^+$ ,  $C_{60}VO^+$ , and  $C_{60}V^+(H_2O)$ . Insets of (a) and (b) show zooming in of the mass range covering only  $C_{60}VO^+$  and  $C_{60}V^+(H_2O)$ , for which the simulated natural isotopic distributions are shown by blue and pink sticks, respectively. (c) IRMPD depletion spectra of  $C_{60}VO^+$  (blue) and  $C_{60}V^+(H_2O)$  (pink) show the concurrent  $C_{60}VO^+$  signal growth and  $C_{60}V^+(H_2O)$  signal loss, as guided by green shading. Figure modified from ref. 3. Copyright 2021 WILEY.

Infrared spectra of the key intermediates with nominal chemical formulas of  $C_{60}V^+(H_2O)$  and  $C_{60}VO^+$  were measured and detailed reaction pathways were modelled with DFT. Figure 8 shows that calculated energy barrier towards  $H_2$  release by  $C_{60}V^+$  is about 60 kJ/mol lower than the barrier starting from isolated  $V^+$ , suggesting a synergy between the metal atom and the  $C_{60}$  support. A remarkable structural phenomenon is observed with  $C_{60}$  behaving as a “hydrogen shuttle”. It temporally retain the transferred hydrogen atom *via* covalent C–H interaction ( $IN1^q$  to  $IN2^t$  via  $TS1^t$ ) and then return it in a barrier-free manner to the metal ( $IN2^t$  to  $IN3^t$  via  $TS2^t$ ). This step facilitates the  $C_{60}VO^+(H_2)$  intermediate formation

(IN4<sup>t</sup>) with an energy barrier of 69 kJ/mol, slightly lower than in the case of VO<sup>+</sup>(H<sub>2</sub>) intermediate formation. Finally, the H<sub>2</sub> is released from the triplet C<sub>60</sub>VO<sup>+</sup> product.



**Figure 8.** Calculated reaction potential energy surfaces (PESs) for (a)  $V^+ + H_2O \rightarrow V^+(H_2O) \rightarrow VO^+ + H_2$  and (b)  $\eta^5 C_{60}V^+ + H_2O \rightarrow C_{60}V^+(H_2O) \rightarrow C_{60}VO^+ + H_2$  at BPW91/6-311++G(2df,2p)//6-31+G(d,p) level. The black and blue lines represent quintet and triplet PESs, respectively. Reactants are labelled RE, reaction intermediates IN, transition states TS, and products P. Green circles indicate schematically the Minimum Energy Crossing Points (MECPs). The labels designated quintet (triplet) states, and the energies are given relative to those of the reactants on the quintet surface. The inset of (b) shows the frontier molecular orbitals (isovalue = 0.04 a.u.) of TS1<sup>t</sup>. Figure adapted from ref. 3. Copyright 2021 WILEY.

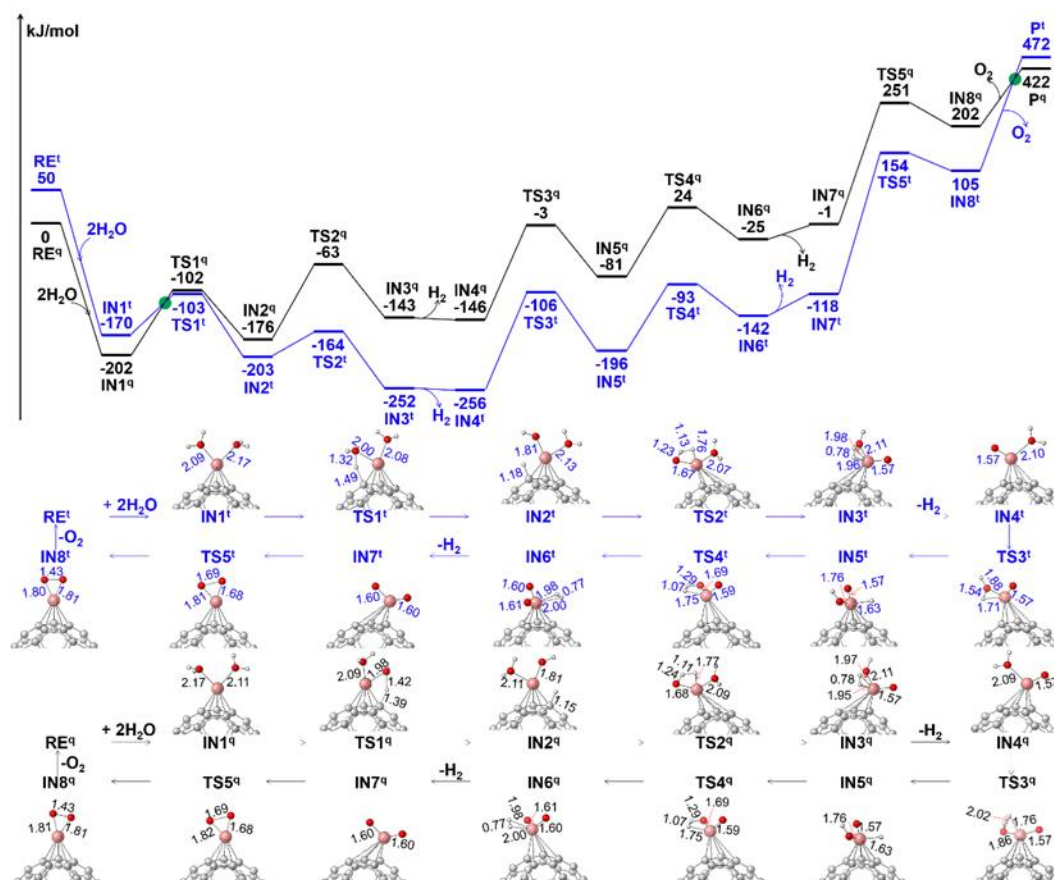
From the frontier molecular orbitals, in particular the highest occupied molecular orbitals of TS1<sup>t</sup> (Figure 8b), the hydrogen transfer can be discerned clearly. The 1s orbital of the transferred hydrogen, the carbon 2p orbital, and the metal d<sub>22</sub> orbital overlap with each other. Such orbital overlap stabilizes TS1, reducing the corresponding energy barrier. The aforementioned insights raise new questions: To what extent does the electronic structure of C<sub>60</sub> contribute to the lowering of the energy barrier? Can replacement of the single metal atom by a few-atom cluster further reduce the barrier? Is it possible to complete the overall catalytic water splitting cycle by C<sub>60</sub>V<sup>+</sup>?



## 5.2 Fullerene as an “electron sponge” for over water splitting: electronic effect

To explore the overall water-splitting performance of  $C_{60}V^+$ , we first calculated the reaction pathways catalyzed by either an isolated  $V^+$  or  $C_{60}$  supported  $V^+$  (Figure 9) following reaction  $2H_2O \xrightarrow{C_{60}V^+} 2H_2 + O_2$ .<sup>4</sup>

Our calculations show that, in the presence of an isolated  $V^+$ , the reaction terminates with  $V^+(OH)_2$  or  $V^+O(H_2O)$ . The two O atoms are too far away to combine. On the contrary, the cycle of overall catalytic water splitting can be completed by  $C_{60}V^+$ , demonstrating the importance of  $C_{60}$ .



**Figure 9.** Calculated reaction pathways for  $2H_2O \rightarrow 2H_2 + O_2$  catalyzed by  $\eta^5 C_{60}V^+$  at BPW91/6-311++G(2df,2p)//6-31G(d,p) level. All the labels follow Figure 8. Figure adapted from ref. 4. Copyright 2022 Elsevier.

The calculated pathways in Figure 9 show that formation of the first  $H_2$  molecule is facilitated by the “hydrogen shuttle” behavior of  $C_{60}$ , followed by transfer of the third H in  $IN4^t$  to the vanadium metal center to form  $IN5^t$ ,  $C_{60}HV^+O(OH)$ , via  $TS3^t$  with an energy barrier of 150 kJ/mol. Subsequently, the fourth H transfers to form  $IN6^t$   $C_{60}(H_2)V^+OO$ , after which a second  $H_2$  is released following the Kubas mechanism.<sup>3,49</sup> This finishes the hydrogen evolution reaction and generates  $IN7^t$ , in which the two O atoms can combine to form  $IN8^t$   $C_{60}V^+(O_2)$ , followed by desorption of  $O_2$  and regeneration of  $C_{60}V^+$ .

Unfortunately,  $O_2$  desorption has a large energy barrier of more than 300 kJ/mol (Figure 9). BOMD simulations under microcanonical ensemble conditions for  $IN8^4$  suggest that before the release of  $O_2$  molecule,  $VO_2^+$  will break up from  $C_{60}V^+(O_2)$ . This raises the question if there exists a suitable metal atom or metal cluster that is as effective as vanadium in dehydrogenating water, while having a much weaker



binding ability of oxygen? Hence, a strategy is needed to either stabilize the metal atom or lower the O<sub>2</sub> desorption barrier.

Examination of the species involved in the overall water splitting show remarkable formal oxidation state changes of the metal center. Natural population analysis indicates that the charges on vanadium and C<sub>60</sub> show reverse trends, suggesting that, C<sub>60</sub> acts as an “electron sponge” in addition to the earlier discussed geometric “hydrogen shuttle” effect. Through accepting and donating electrons in different steps, the oxidation state of the metal center changes along the reaction pathway.<sup>4</sup> Such mechanism could apply to other C<sub>60</sub>-supported metal catalysts as well. For example, Xie and co-workers recently reported that C<sub>60</sub>-buffered Cu/SiO<sub>2</sub> catalysts facilitate ethylene glycol (EG) synthesis from dimethyl oxalate (DMO) at ambient pressure, and proposed that the “electron buffer” effect of C<sub>60</sub> is crucial in stabilizing the active Cu<sup>+</sup> sites.<sup>50</sup> It should be noted that the “electron sponge” and the “electron buffer” effects are conceptually the same. The idea of using C<sub>60</sub> to prompt redox reactions has been realized in several catalytic processes, including hydrogen evolution reactions<sup>4,51-53</sup> and CO<sub>2</sub> reduction.<sup>54</sup>

## 6. CONCLUSIONS AND PERSPECTIVES

In view of brevity, we presented in this account only part of our work on fullerene-metal clusters. We also studied hydrogen adsorption on fullerene supported cobalt clusters, and the size dependent reactivity of fullerene supported vanadium cationic clusters towards water splitting. The techniques to synthesize fullerene-metals and to measure their vibrational spectra using action spectroscopy, allow to identify the binding positions of metal atoms and clusters on the fullerene, their chemical bonding, structural evolution, and support effects on the structure, with the help of theoretical calculations. Explorations of those fundamental aspects in term of structures and chemical bonding will continue.

The synthesis and spectroscopic approaches are not limited to fullerene-metal cationic clusters. In principle, we can synthesize any composite fullerene clusters, for example fullerene-nonmetal clusters, fullerene-organic molecular clusters, also in neutral or negatively charged states. In our opinion, more exciting aspects of hybrid fullerene clusters are awaiting exploration, for instance:

**Fullerene in space:** Looking at the sky, where are we from and what’s the origin of life? Those philosophical questions keep inspiring human beings. Our proposal on the potential existence of fullerene-metal complexes in space will stimulate astronomers and spectroscopists to search for them in space and obtain more laboratory evidence to prove or refute it. Their identification can potentially link chemical pathways and formation mechanisms of various carbonaceous species and thus carbon chemistry in space. On one hand, measuring the absorption spectra of fullerene-metal clusters in the ultraviolet-visible spectral range is expected to attract great attention given their potential link to the century mystery about the carriers of DIBs. On the other hand, fullerene-organic molecular clusters, in particular those with aromatic hydrocarbons, deserve investigation since both fullerene and aromatic hydrocarbons have been found in astrophysical conditions and their cycloaddition reactions are facile.

**Fullerene in catalysis:** The idea of utilizing the “electron sponge” effect of fullerenes in catalysis has quickly gained attention from peer researchers and permit potential utilization in hydrogen technologies. The remarkable geometric and electronic effects of C<sub>60</sub> are now investigated in more catalytic reactions. We are particularly interested in exploring the effect of metal type, cluster composition, and cluster size, with the purpose to gain deeper insight in the structure-reactivity relationship, structural fluxionality, and metal-support interactions. These studies are expected to provide guidance for designing high-

performance dehydrogenation catalysts that can make efficient use of scarce raw materials and produce green hydrogen or other precious chemicals.

Furthermore, the electron sponge effect of fullerene in catalysis motivates a detailed investigation of charge-transfer dynamics between metal and C<sub>60</sub>. This could provide understanding about how and on which time scale the electron transfer happens. Such information is of significance to design hybrid fullerene-involved junctions and structures for nanoscale electronic devices and photovoltaic materials. In addition, the recent successful synthesis of 2D fullerene network<sup>55</sup> motivates exploration of the synthesis of 2D metal-doped fullerene networks, because metal doping induces interesting electronic, optical, and magnetic properties.

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### Notes

The authors declare no competing financial interests.

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## REFERENCES

- (1) Xu, J. Z.; Bakker, J. M.; Lushchikova, O. V.; Lievens, P.; Janssens, E.; Hou, G.-L. Pentagon, Hexagon, or Bridge? Identifying the Location of a Single Vanadium Cation on Buckminsterfullerene Surface. *J. Am. Chem. Soc.* **2023**, *145*, 22243-22251.
- (2) Hou, G.-L.; Lushchikova, O. V.; Bakker, J. M.; Lievens, P.; Decin, L.; Janssens, E. Buckyball-metal complexes as potential carriers of astronomical unidentified infrared emission bands. *Astrophys. J.* **2023**, *952*, 13.
- (3) Hou, G.-L.; Yang, T.; Li, M.; Vanbuel, J.; Lushchikova, O. V.; Ferrari, P.; Bakker, J. M.; Janssens, E. Water splitting by C<sub>60</sub>-supported vanadium single atoms. *Angew. Chem. Int. Ed.* **2021**, *60*, 27095-27101.
- (4) Li, M.; Yang, T.; Bakker, J. M.; Janssens, E.; Hou, G.-L. Unveiling the role of C<sub>60</sub>-supported vanadium single atoms for catalytic overall water splitting. *Cell Rep. Phys. Sci.* **2022**, *3*, 100910.
- (5) Kroto, H. W.; Heath, J. R.; O'Brien, S. C.; Curl, R. F.; Smalley, R. E. C<sub>60</sub>: Buckminsterfullerene. *Nature* **1985**, *318*, 162-163.
- (6) Curl, R. F.; Smalley, R. E. Probing C<sub>60</sub>. *Science* **1988**, *242*, 1017-1022.
- (7) Kroto, H. W.; Allaf, A.; Balm, S. C<sub>60</sub>: Buckminsterfullerene. *Chem. Rev.* **1991**, *91*, 1213-1235.
- (8) Krätschmer, W.; Lamb, L. D.; Fostiropoulos, K.; Huffman, D. R. Solid C<sub>60</sub>: a new form of carbon. *Nature* **1990**, *347*, 354-358.
- (9) Heath, J.; O'Brien, S.; Zhang, Q.; Liu, Y.; Curl, R.; Tittel, F.; Smalley, R. Lanthanum complexes of spheroidal carbon shells. *J. Am. Chem. Soc.* **1985**, *107*, 7779-7780.
- (10) Fagan, P. J.; Calabrese, J. C.; Malone, B. Metal complexes of buckminsterfullerene (C<sub>60</sub>). *Acc. Chem. Res.* **1992**, *25*, 134-142.
- (11) Popov, A. A.; Yang, S.; Dunsch, L. Endohedral fullerenes. *Chem. Rev.* **2013**, *113*, 5989-6113.
- (12) Lebedeva, M. A.; Chamberlain, T. W.; Khlobystov, A. N. Harnessing the synergistic and complementary properties of fullerene and transition-metal compounds for nanomaterial applications. *Chem. Rev.* **2015**, *115*, 11301-11351.
- (13) Chandrakumar, K.; Ghosh, S. K. Alkali-metal-induced enhancement of hydrogen adsorption in C<sub>60</sub> fullerene: an ab initio study. *Nano Lett.* **2008**, *8*, 13-19.
- (14) Yoon, M.; Yang, S.; Hicke, C.; Wang, E.; Geohegan, D.; Zhang, Z. Calcium as the superior coating metal in functionalization of carbon fullerenes for high-capacity hydrogen storage. *Phys. Rev. Lett.* **2008**, *100*, 206806.
- (15) Stephens, P. W.; Mihaly, L.; Lee, P. L.; Whetten, R. L.; Huang, S.-M.; Kaner, R.; Deiderich, F.; Holczer, K. Structure of single-phase superconducting K<sub>3</sub>C<sub>60</sub>. *Nature* **1991**, *351*, 632-634.
- (16) Chen, Y. C.; Hsu, C. Y.; Lin, R. Y. Y.; Ho, K. C.; Lin, J. T. Materials for the active layer of organic photovoltaics: ternary solar cell approach. *ChemSusChem* **2013**, *6*, 20-35.
- (17) Park, H.; Park, J.; Lim, A. K.; Anderson, E. H.; Alivisatos, A. P.; McEuen, P. L. Nanomechanical oscillations in a single-C<sub>60</sub> transistor. *Nature* **2000**, *407*, 57-60.
- (18) Yee, S. K.; Malen, J. A.; Majumdar, A.; Segalman, R. A. Thermoelectricity in fullerene-metal heterojunctions. *Nano Lett.* **2011**, *11*, 4089-4094.

- (19) Shukla, M. K.; Dubey, M.; Leszczynski, J. Theoretical Investigation of Electronic Structures and Properties of C<sub>60</sub>- Gold Nanocontacts. *ACS Nano* **2008**, *2*, 227-234.
- (20) De, D. S.; Flores-Livas, J. A.; Saha, S.; Genovese, L.; Goedecker, S. Stable structures of exohedrally decorated C<sub>60</sub>-fullerenes. *Carbon* **2018**, *129*, 847-853.
- (21) Kroto, H. W.; Jura, M. Circumstellar and interstellar fullerenes and their analogues. *Astron. Astrophys.* **1992**, *263*, 275-280.
- (22) Cami, J.; Bernard-Salas, J.; Peeters, E.; Malek, S. E. Detection of C<sub>60</sub> and C<sub>70</sub> in a young planetary nebula. *Science* **2010**, *329*, 1180-1182.
- (23) Woods, P. The discovery of cosmic fullerenes. *Nat. Astron.* **2020**, *4*, 299-305.
- (24) Campbell, E. K.; Holz, M.; Gerlich, D.; Maier, J. P. Laboratory confirmation of C<sub>60</sub><sup>+</sup> as the carrier of two diffuse interstellar bands. *Nature* **2015**, *523*, 322-323.
- (25) Cordiner, M.; Linnartz, H.; Cox, N.; Cami, J.; Najarro, F.; Proffitt, C.; Lallement, R.; Ehrenfreund, P.; Foing, B.; Gull, T. Confirming interstellar C<sub>60</sub><sup>+</sup> using the hubble space telescope. *Astrophys. J. Lett.* **2019**, *875*, L28.
- (26) Kwok, S. The mystery of unidentified infrared emission bands. *Astrophys. Space Sci.* **2022**, *367*, 16.
- (27) Linnartz, H.; Cami, J.; Cordiner, M.; Cox, N.; Ehrenfreund, P.; Foing, B.; Gatchell, M.; Scheier, P. C<sub>60</sub><sup>+</sup> as a diffuse interstellar band carrier; a spectroscopic story in 6 acts. *J. Mol. Spectrosc.* **2020**, *367*, 111243.
- (28) Hou, G.-L.; Janssens, E. Supported and Isolated Metal Atoms and Clusters as Models for Understanding the Hydrogen Economy. *Fundam. Res.* **2023**, DOI: 10.1016/j.fmre.2023.1010.1011.
- (29) Le, H. T.; Lang, S. M.; De Haeck, J.; Lievens, P.; Janssens, E. Carbon monoxide adsorption on neutral and cationic vanadium doped gold clusters. *Phys. Chem. Chem. Phys.* **2012**, *14*, 9350-9358.
- (30) Fielicke, A. Probing the binding and activation of small molecules by gas-phase transition metal clusters via IR spectroscopy. *Chem. Soc. Rev.* **2023**, *52*, 3778-3841.
- (31) Bakker, J. M.; Lapoutre, V. J.; Redlich, B.; Oomens, J.; Sartakov, B. G.; Fielicke, A.; Von Helden, G.; Meijer, G.; Van Der Meer, A. F. Intensity-resolved IR multiple photon ionization and fragmentation of C<sub>60</sub>. *J. Chem. Phys.* **2010**, *132*, 074305.
- (32) Fu, W.; Hopkins, W. S. Applying machine learning to vibrational spectroscopy. *J. Phys. Chem. A* **2018**, *122*, 167-171.
- (33) Robledo, M.; Aguirre, N. F.; Díaz-Tendero, S.; Martín, F.; Alcamí, M. Bonding in exohedral metal-fullerene cationic complexes. *RSC Adv.* **2014**, *4*, 53010-53020.
- (34) Nagao, S.; Kurikawa, T.; Miyajima, K.; Nakajima, A.; Kaya, K. Formation and Structures of Transition Metal-C<sub>60</sub> Clusters. *J. Phys. Chem. A* **1998**, *102*, 4495-4500.
- (35) German, E.; Hou, G.-L.; Vanbuel, J.; Bakker, J. M.; Alonso, J. A.; Janssens, E.; López, M. J. Infrared spectra and structures of C<sub>60</sub>Rh<sub>n</sub><sup>+</sup> complexes. *Carbon* **2022**, *197*, 535-543.
- (36) Harding, D. J.; Gruene, P.; Haertelt, M.; Meijer, G.; Fielicke, A.; Hamilton, S. M.; Hopkins, W. S.; Mackenzie, S. R.; Neville, S.; Walsh, T. R. Probing the structures of gas-phase rhodium cluster cations by far-infrared spectroscopy. *J. Chem. Phys.* **2010**, *133*, 214304.
- (37) Vanbuel, J.; Germán, E.; Libeert, G.; Veys, K.; Moens, J.; Alonso, J. A.; López, M. J.; Janssens, E. Reactivity of Cobalt-Fullerene Complexes towards Deuterium. *ChemPhysChem* **2020**, *21*, 1012-1018.
- (38) Xu, J.; Hou, G.-L. Comment on "Multiple locations of boron atoms in the exohedral and endohedral C<sub>60</sub> fullerene". *Phys. Rev. A* **2022**, *106*, 056801.
- (39) Kwok, S. Complex organics in space from Solar System to distant galaxies. *Astron. Astrophys. Rev.* **2016**, *24*, 8.
- (40) McGuire, B. A. 2021 census of interstellar, circumstellar, extragalactic, protoplanetary disk, and exoplanetary molecules. *Astrophys. J. Suppl. Ser.* **2022**, *259*, 30.
- (41) Highberger, J.; Savage, C.; Bieging, J.; Ziurys, L. Heavy-metal chemistry in proto-planetary nebulae: Detection of MgNC, NaCN, and AlF toward CRL 2688. *Astrophys. J.* **2001**, *562*, 790.

- (42) Omont, A. Interstellar fullerene compounds and diffuse interstellar bands. *Astron. Astrophys.* **2016**, *590*, A52.
- (43) Calvo, F.; Li, Y.; Kiawi, D. M.; Bakker, J. M.; Parneix, P.; Janssens, E. Nonlinear effects in infrared action spectroscopy of silicon and vanadium oxide clusters: experiment and kinetic modeling. *Phys. Chem. Chem. Phys.* **2015**, *17*, 25956-25967.
- (44) Dunk, P. W.; Adjizian, J.-J.; Kaiser, N. K.; Quinn, J. P.; Blakney, G. T.; Ewels, C. P.; Marshall, A. G.; Kroto, H. W. Metallofullerene and fullerene formation from condensing carbon gas under conditions of stellar outflows and implication to stardust. *Proc. Natl. Acad. Sci. U.S.A* **2013**, *110*, 18081-18086.
- (45) Dunk, P. W.; Mulet-Gas, M.; Nakanishi, Y.; Kaiser, N. K.; Rodríguez-Forte, A.; Shinohara, H.; Poblet, J. M.; Marshall, A. G.; Kroto, H. W. Bottom-up formation of endohedral mono-metallofullerenes is directed by charge transfer. *Nat. Commun.* **2014**, *5*, 5844.
- (46) Xu, J. Z.; Li, A. G.; Li, X. H.; Hou, G.-L. Towards a reliable prediction of the infrared spectra of cosmic fullerenes and their derivatives in the JWST era. *Mon. Notices Royal Astron. Soc.* **2023**, *525*, 3061-3074.
- (47) Yang, X.-F.; Wang, A.; Qiao, B.; Li, J.; Liu, J.; Zhang, T. Single-atom catalysts: a new frontier in heterogeneous catalysis. *Acc. Chem. Res.* **2013**, *46*, 1740-1748.
- (48) Carnegie, P. D.; Marks, J. H.; Brathwaite, A. D.; Ward, T. B.; Duncan, M. A. Microsolvation in  $V^+(H_2O)_n$  clusters studied with selected-ion infrared spectroscopy. *J. Phys. Chem. A* **2020**, *124*, 1093-1103.
- (49) Kubas, G. J. Fundamentals of  $H_2$  binding and reactivity on transition metals underlying hydrogenase function and  $H_2$  production and storage. *Chemical Reviews* **2007**, *107*, 4152-4205.
- (50) Zheng, J.; Huang, L.; Cui, C. H.; Chen, Z. C.; Liu, X. F.; Duan, X.; Cao, X. Y.; Yang, T. Z.; Zhu, H.; Shi, K.; Du, P.; Ying, S. W.; Zhu, C. F.; Yao, Y. G.; Guo, G. C.; Yuan, Y.; Xie, S. Y.; Zheng, L. S. Ambient-pressure synthesis of ethylene glycol catalyzed by  $C_{60}$ -buffered Cu/SiO<sub>2</sub>. *Science* **2022**, *376*, 288-292.
- (51) Zhang, R.; Li, Y.; Zhou, X.; Yu, A.; Huang, Q.; Xu, T.; Zhu, L.; Peng, P.; Song, S.; Echegoyen, L. Single-atomic platinum on fullerene  $C_{60}$  surfaces for accelerated alkaline hydrogen evolution. *Nat. Commun.* **2023**, *14*, 2460.
- (52) Luo, T.; Huang, J.; Hu, Y.; Yuan, C.; Chen, J.; Cao, L.; Kajiyoshi, K.; Liu, Y.; Zhao, Y.; Li, Z.; Feng, Y. Fullerene lattice-confined Ru nanoparticles and single atoms synergistically boost electrocatalytic hydrogen evolution reaction. *Adv. Funct. Mater.* **2023**, *33*, 2213058.
- (53) Chen, J.; Aliasgar, M.; Zamudio, F. B.; Zhang, T.; Zhao, Y.; Lian, X.; Wen, L.; Yang, H.; Sun, W.; Kozlov, S. M.; Chen, W.; Wang, L. Diversity of platinum-sites at platinum/fullerene interface accelerates alkaline hydrogen evolution. *Nat. Commun.* **2023**, *14*, 1711.
- (54) Zhao, B.; Chen, F.; Cheng, C.; Li, L.; Liu, C.; Zhang, B.  $C_{60}$ -stabilized  $Cu^+$  sites boost electrocatalytic reduction of  $CO_2$  to  $C_2^+$  products. *Adv. Energy Mater.* **2023**, *13*, 2204346.
- (55) Hou, L.; Cui, X.; Guan, B.; Wang, S.; Li, R.; Liu, Y.; Zhu, D.; Zheng, J. Synthesis of a monolayer fullerene network. *Nature* **2022**, *606*, 507-510.